Week 3

Class Activity 6

Group Activity: Activity: Exploring ε in Differential Privacy (Laplace Mechanism)

Scenario: A survey of 100 people found 20 smoke.
We will apply Laplace noise to protect privacy.
Noise is added according to ε (privacy level).
The smaller ε → the *stronger* the privacy → the *larger* the noise.

Instructions:

1. Use the table below.
2. For each ε value, choose one random noise number from the given range.
3. Compute:
   New Count = 20 + noise
4. Discuss: Which results look realistic, and which are too distorted to be useful?

Noise Ranges

ε	Privacy Strength	Accuracy	Choose Noise From Range	Example Calculation
5.0			**–0.5 to +0.5**	
1.0			**–2 to +2**	
0.5			**–4 to +4**	
0.1			**–10 to +10**	
0.01			**–100 to +100**	